Project Title: Technology Trends in the Developer Job Market

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You can find all of the presentation and more materials at
https://github.com/pmntang/Project-for-Data-Science

OUTLINE

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- **Results**
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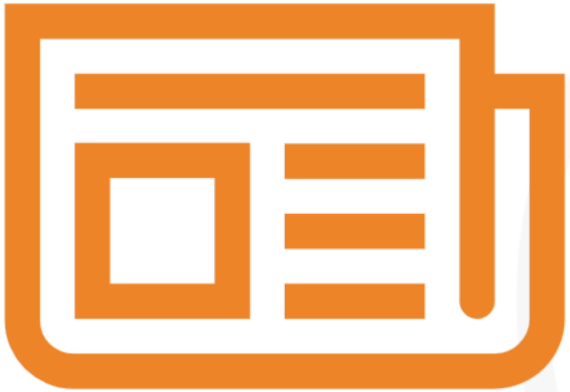
EXECUTIVE SUMMARY

- **Python and JavaScript** dominate current job market demand and future developer interest.
- **SQL and PostgreSQL** are critical database skills, showing high current usage and strong future demand.
- **Cloud platforms, especially AWS,** are the most desired technology for future projects.
- The developer demographic is predominantly **25-34 years old**, holding a Bachelor's degree.
- High salaries for **Swift and C++** indicate premium value for niche and performance-critical skills.

INTRODUCTION

- **Purpose:** To analyze current and emerging trends in programming languages and databases to guide career and learning paths for aspiring data analysts and developers.
- **Target Audience:** Job seekers, career switchers, educational institutions, and corporate L&D departments.
- **Value:** Provides data-driven insights into which technologies are in high demand and offer the best financial returns, enabling informed decision-making.

METHODOLOGY



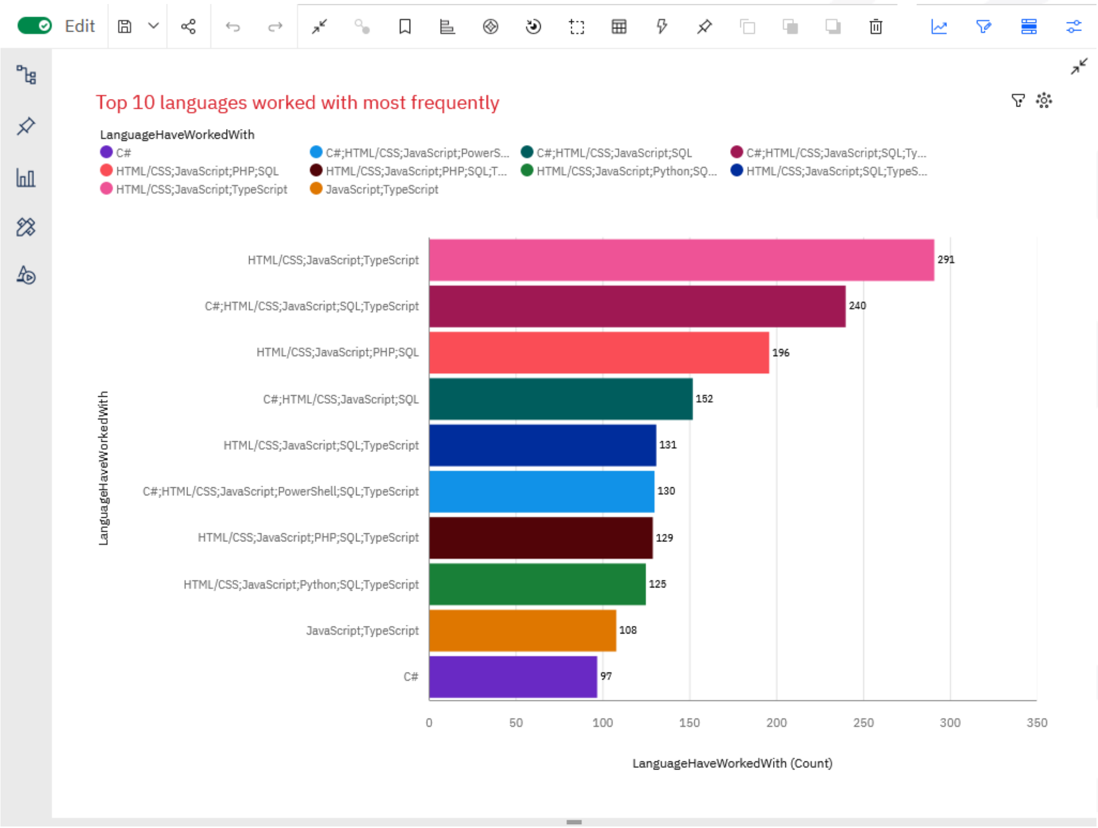
- **Data Sources:**
  - Stack Overflow Developer Survey (for current usage and future desire).
  - Job Postings API (job_postings.xlsx).
  - Web Scraping of salary data (popular-languages.xlsx).

- **Collection Methods:** Automated API calls, web scraping, and survey data aggregation.

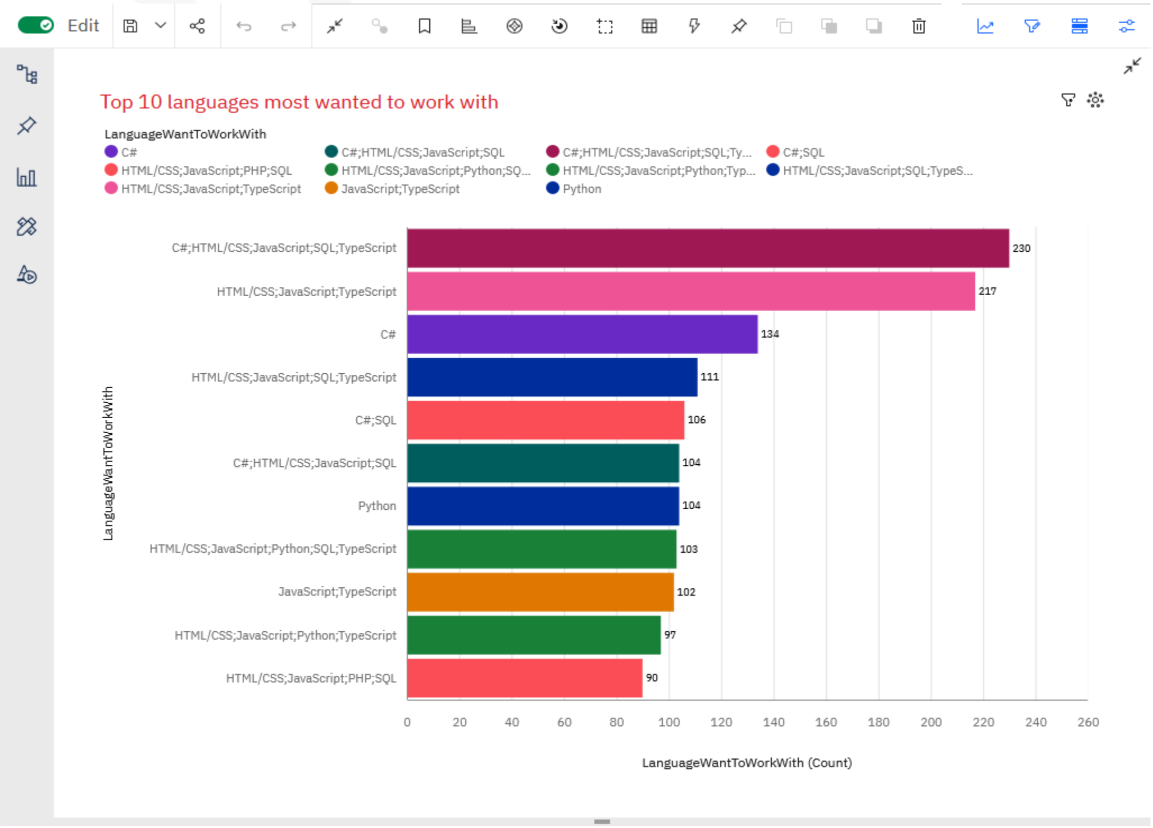
- **Key Wrangling Steps:** Data cleaning, handling missing values, aggregating counts, and calculating percentages for comparative analysis.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Years



Beyond the C# language itself, which is a worthwhile investment to learn and master, one should consider specializing in

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The web stack, HTML, CSS, JavaScript (TypeScript), and PHP (to a lesser extent) continue to dominate in terms of the most widely used languages or combinations of languages. It is worth noting the presence of SQL and Python among the most sought-after language combinations
- C# stands alone as the most widely used traditional language at present; no doubt this has something to do with the dominance of Microsoft hard/software and its cloud infrastructure.

Implications

- Training should emphasize the web stack HTML, CSS, JavaScript (TypeScript), PHP (to a lesser extent) and Python, SQL for employability.
- Beyond the C# language itself, which is a worthwhile investment to learn and master, one should consider specializing in cloud computing to ensure employability.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

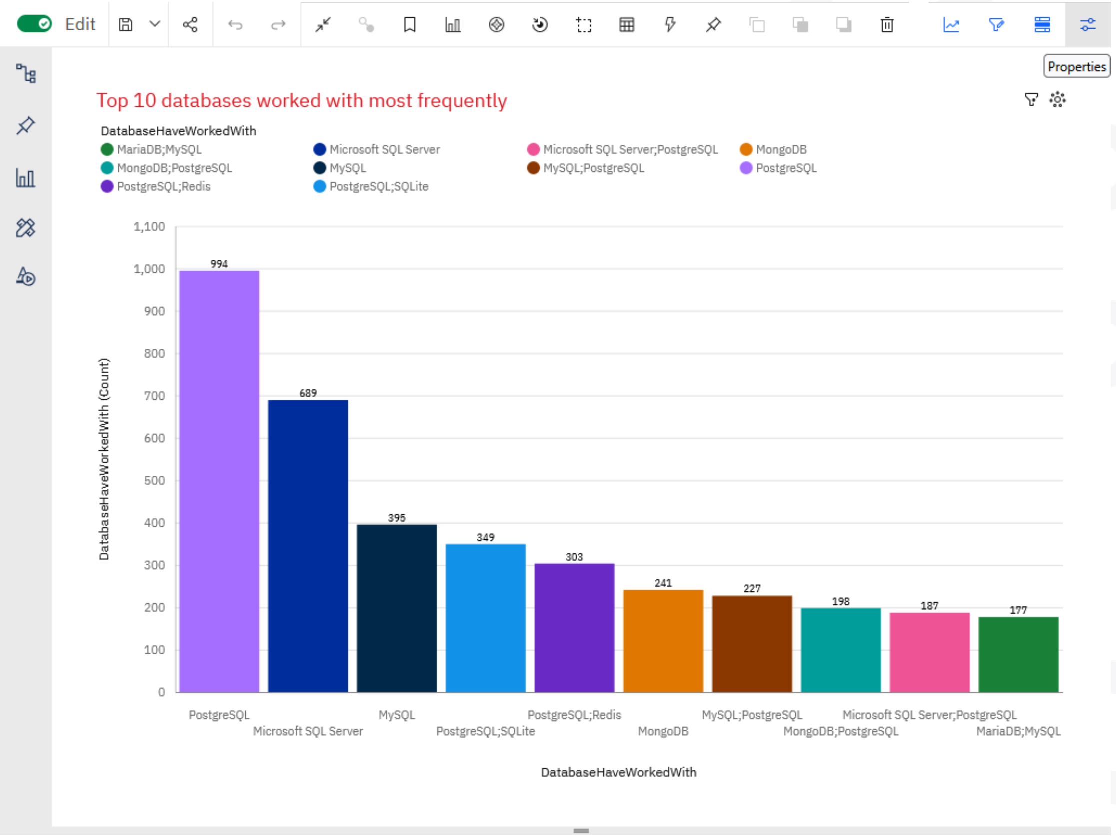
- The trend looks set to continue in the future, but with one nuance: Python will become more popular, while C# and SQL will be increasingly in demand. The PowerShell scripting language is expected to disappear from the top 10 most in-demand languages; is this a result of the widespread adoption of the cloud?

Implications

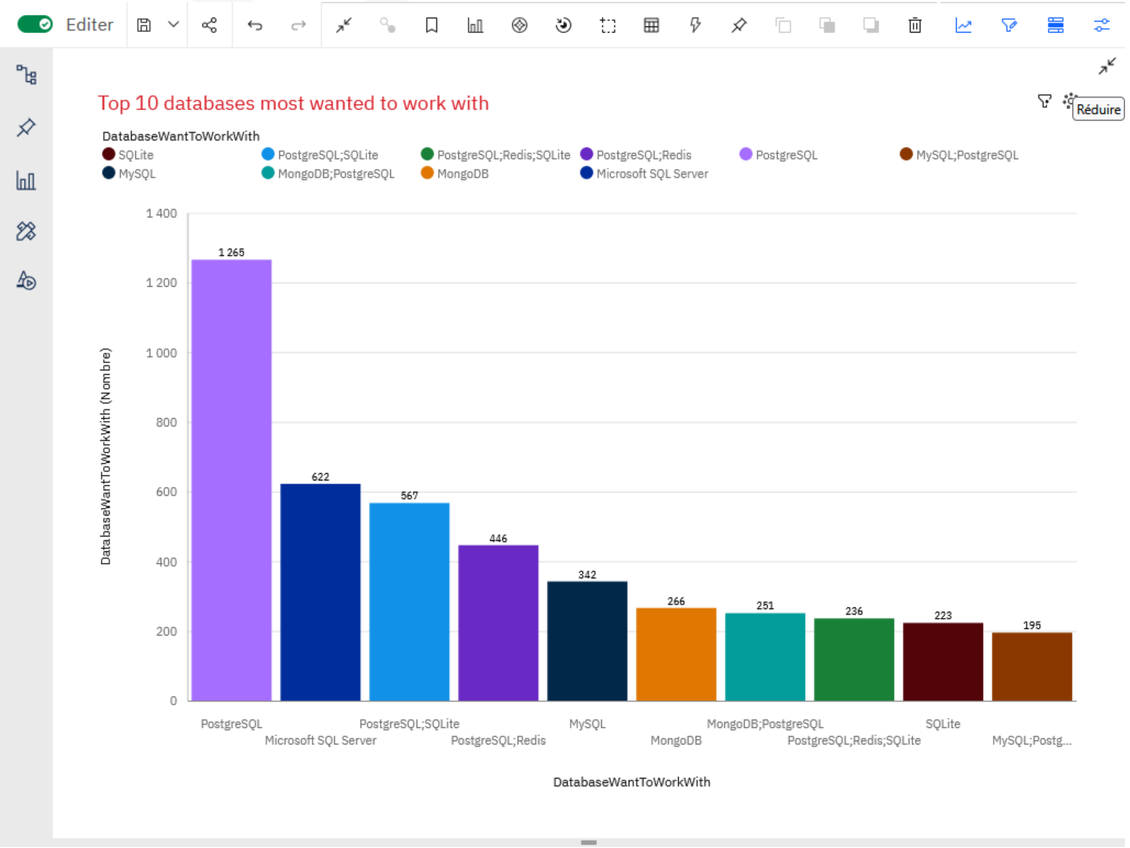
- The cloud is no longer just an option, but a major trend, as evidenced by the decline in non-mandatory languages in cloud usage. On the other hand, languages considered essential for the cloud will be increasingly in demand. It is therefore crucial to invest in these languages.

DATABASE TRENDS

Current Year



Next Years



DATABASE TRENDS - FINDINGS & IMPLICATIONS

- Using dashboard insights, relational databases remain dominant.
- Cloud-hosted and NoSQL options continue to grow with application scale.
- Future demand points to managed, scalable database services.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Traditional SQL databases are still widely used.
- Future-facing roles expect familiarity with cloud DBs.
- We see SQLite's role growing stronger in the future, which is both an indicator of diversification and a shift toward the cloud, since it is more of a cross-functional SQL tool in a cloud environment than in a traditional relational database management environment.

Implications

- Include SQL in all core training tracks.
- Add modules on cloud-native databases.
- Skill diversification improves job readiness. It is therefore advisable to rely on traditional SQL technologies to seek expertise in emerging but already essential technologies such as the Cloud.

DASHBOARD

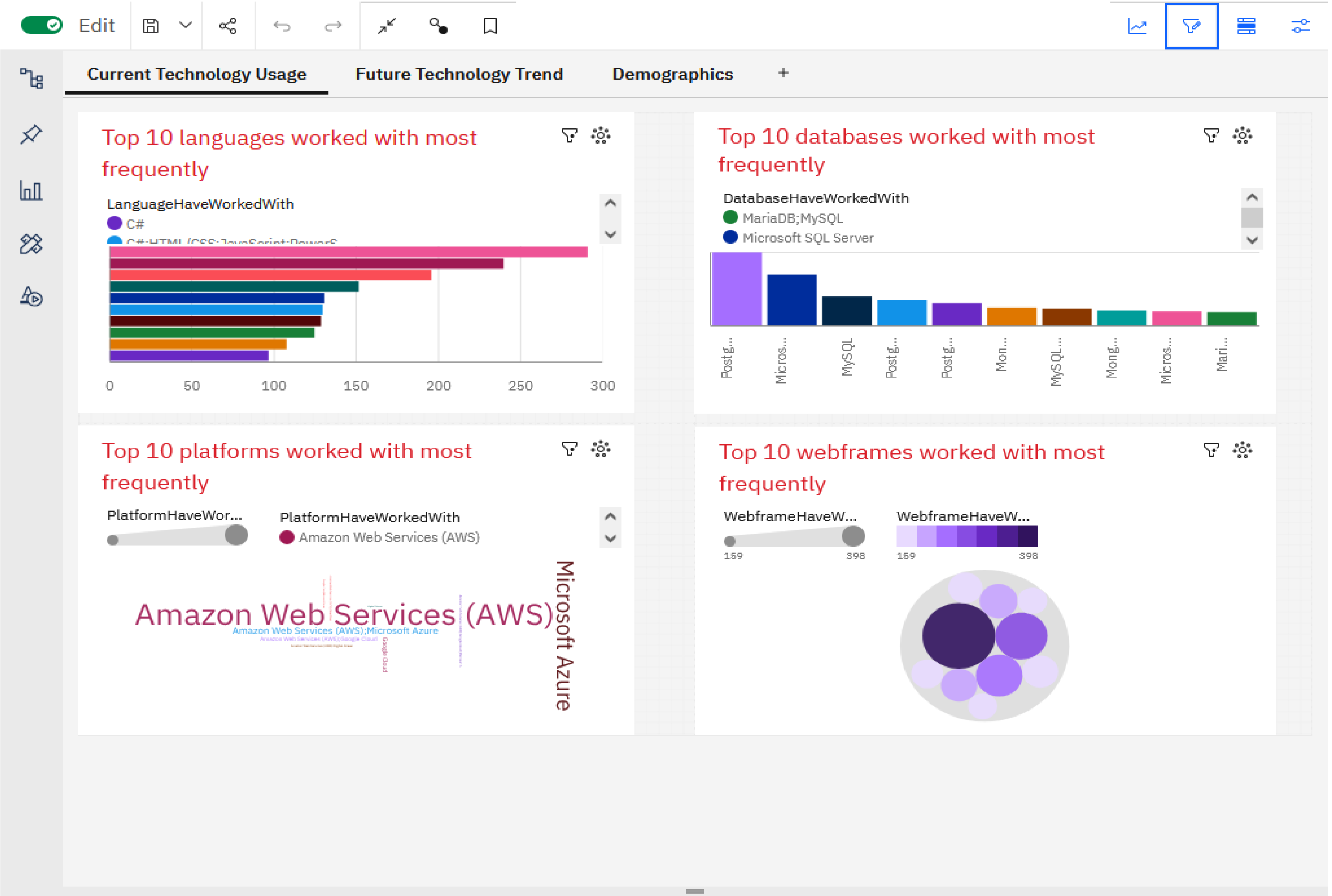
- The following dashboards were created in Cognos Analytics:

- Current Technology Usage

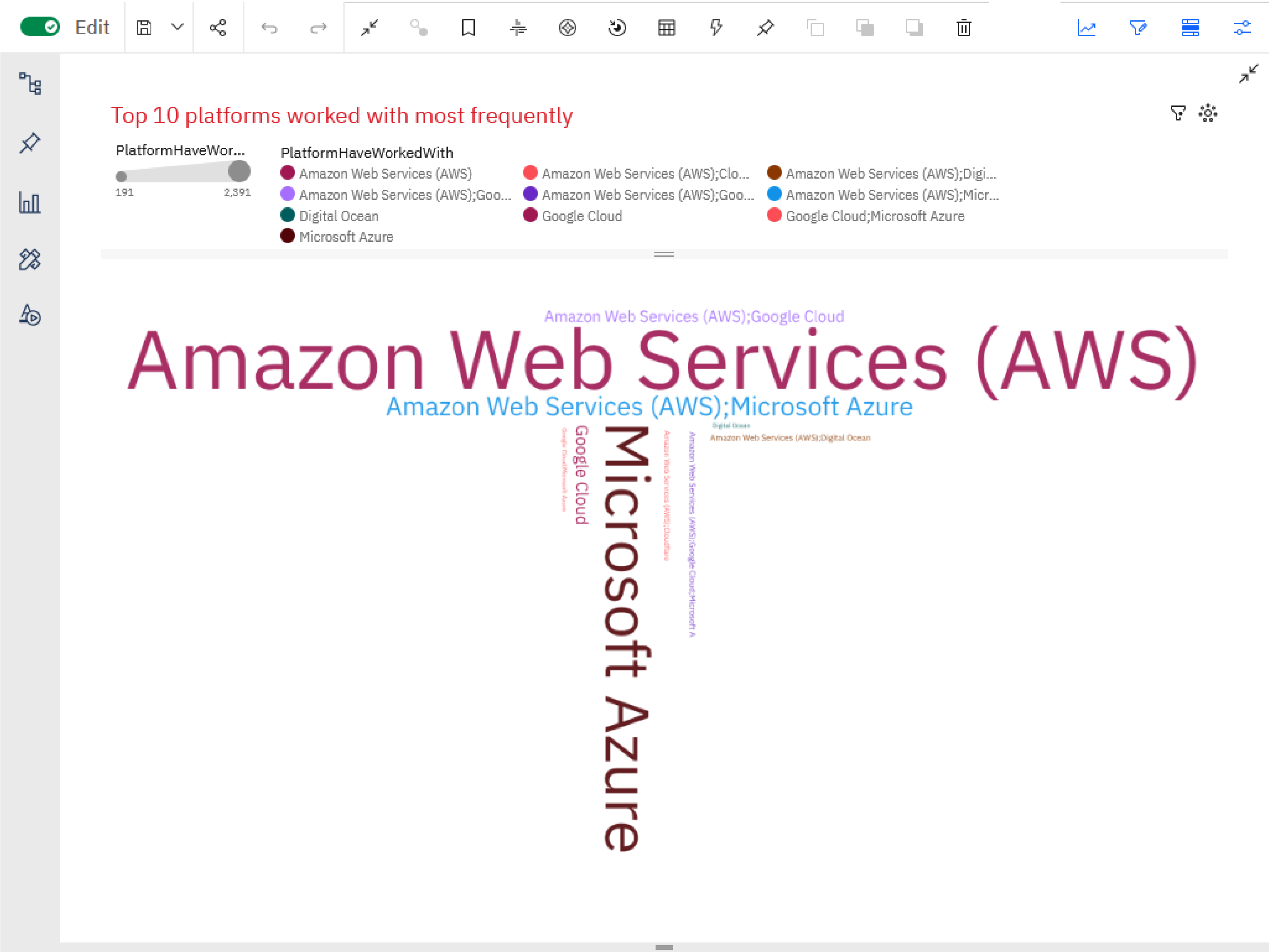
- Future Technology Trends

- Demographics

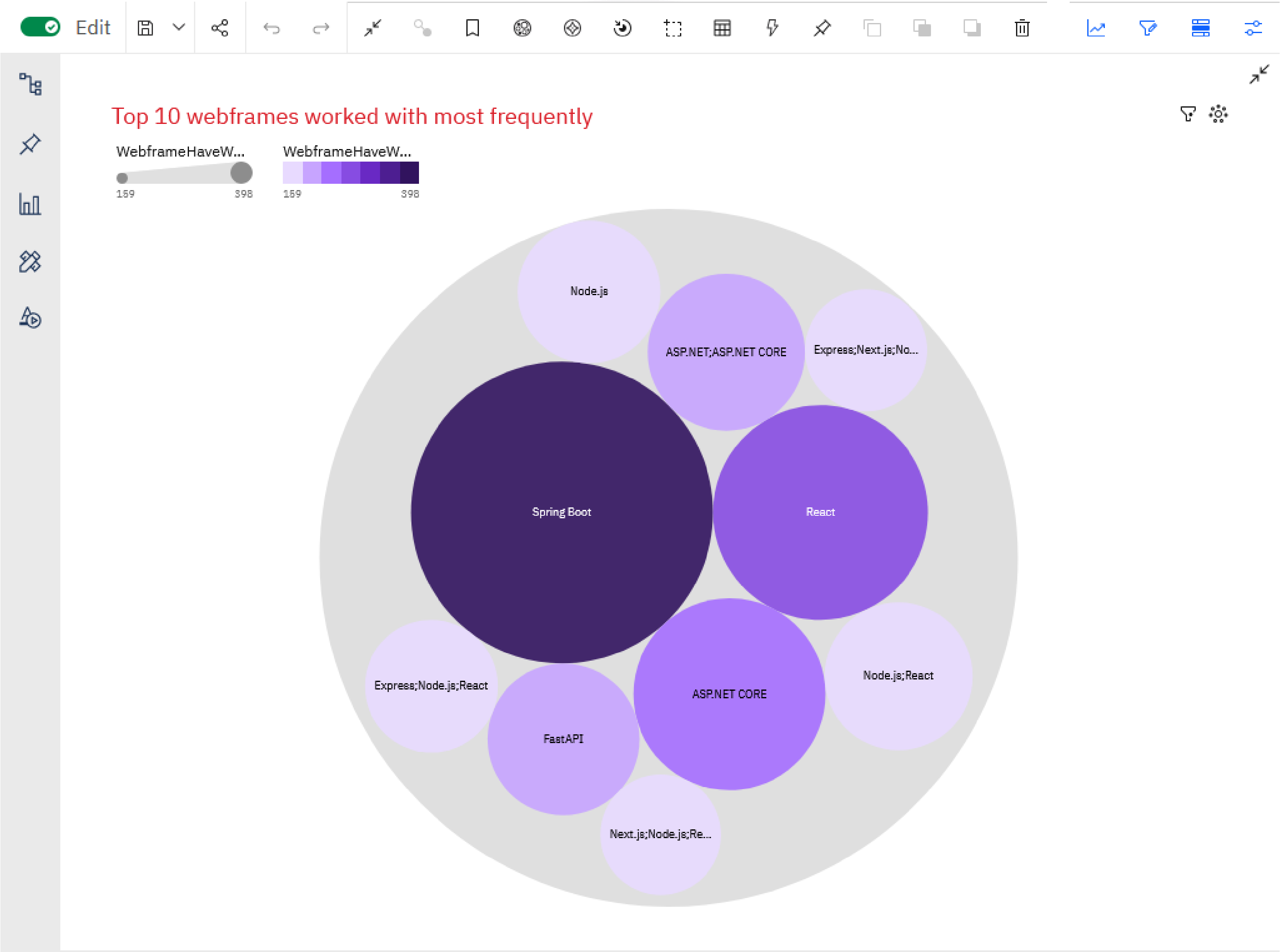
Current Technology Usage



Current Technology Usage -Top 10 platforms worked with most frequently

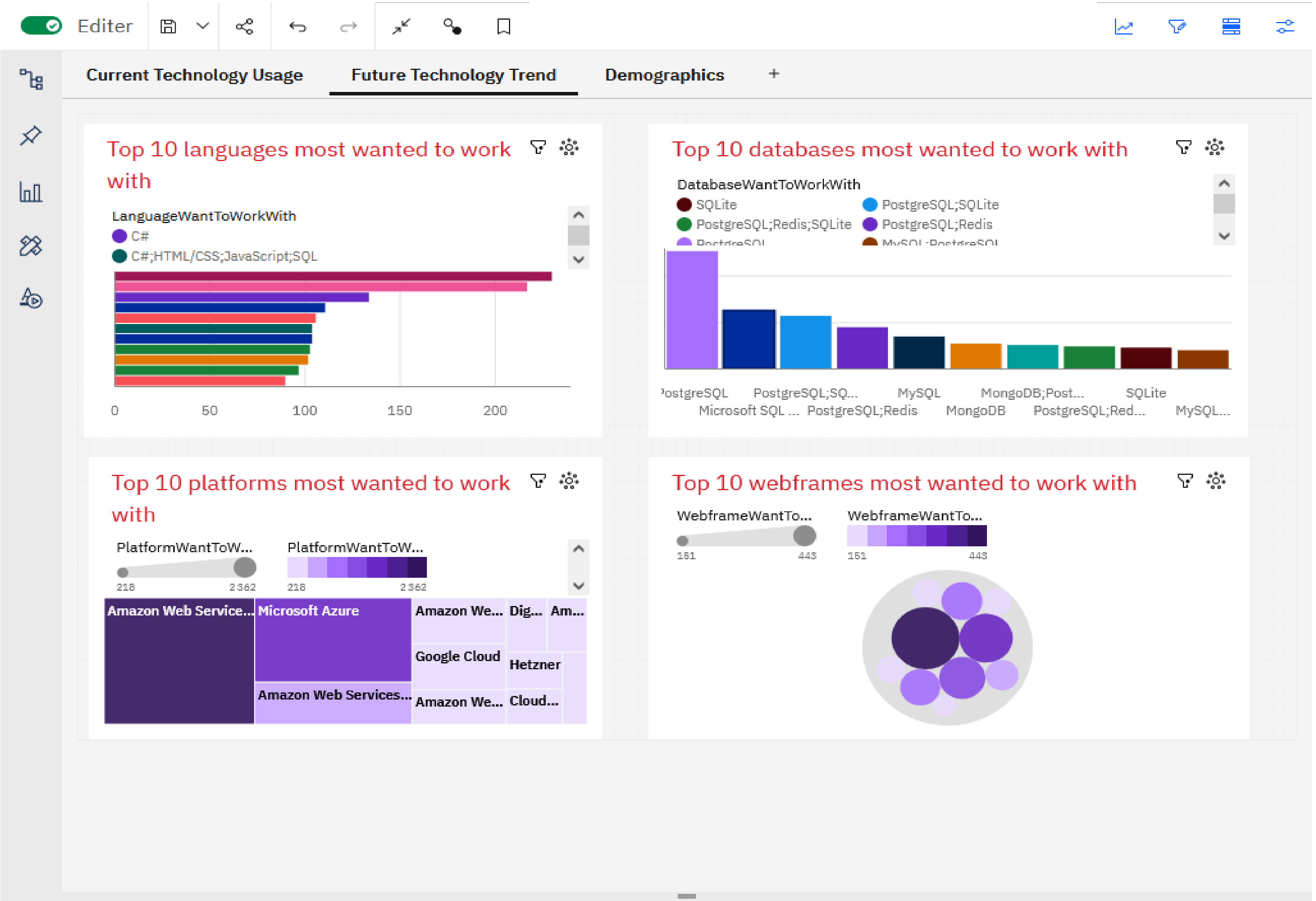


Current Technology Usage - Top 10 webframes worked with most frequently

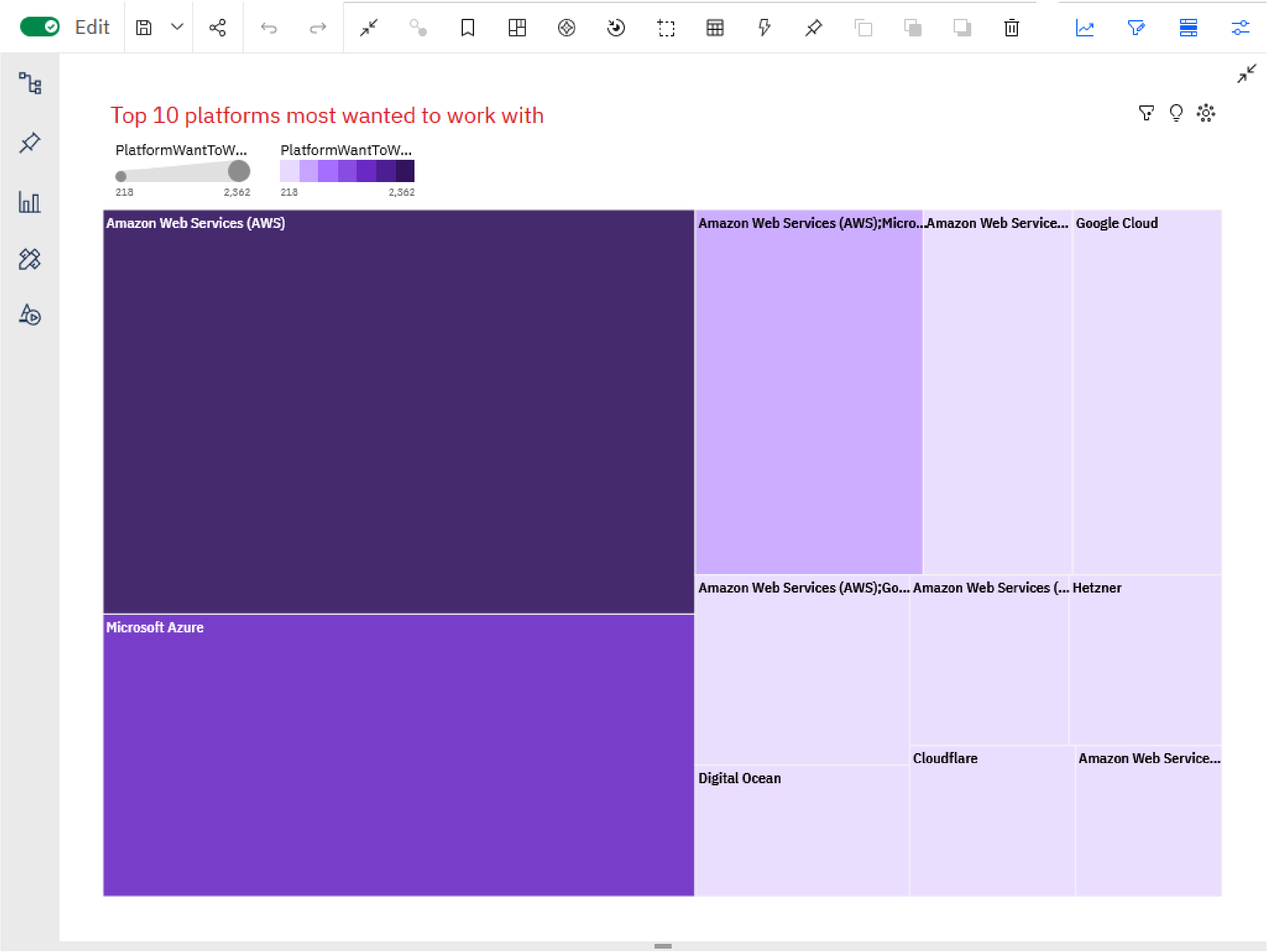


Skills Network

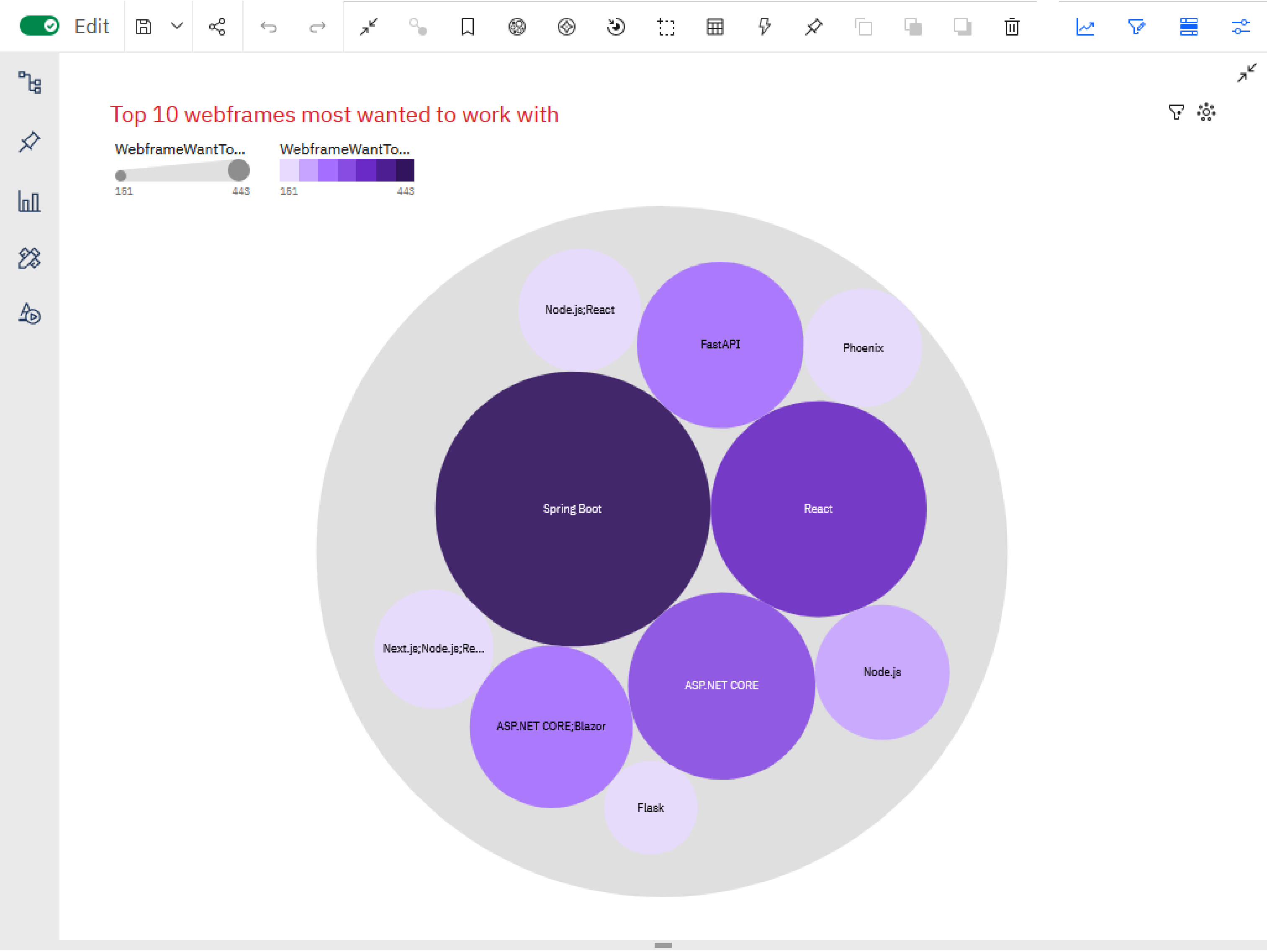
Future Technology Trend



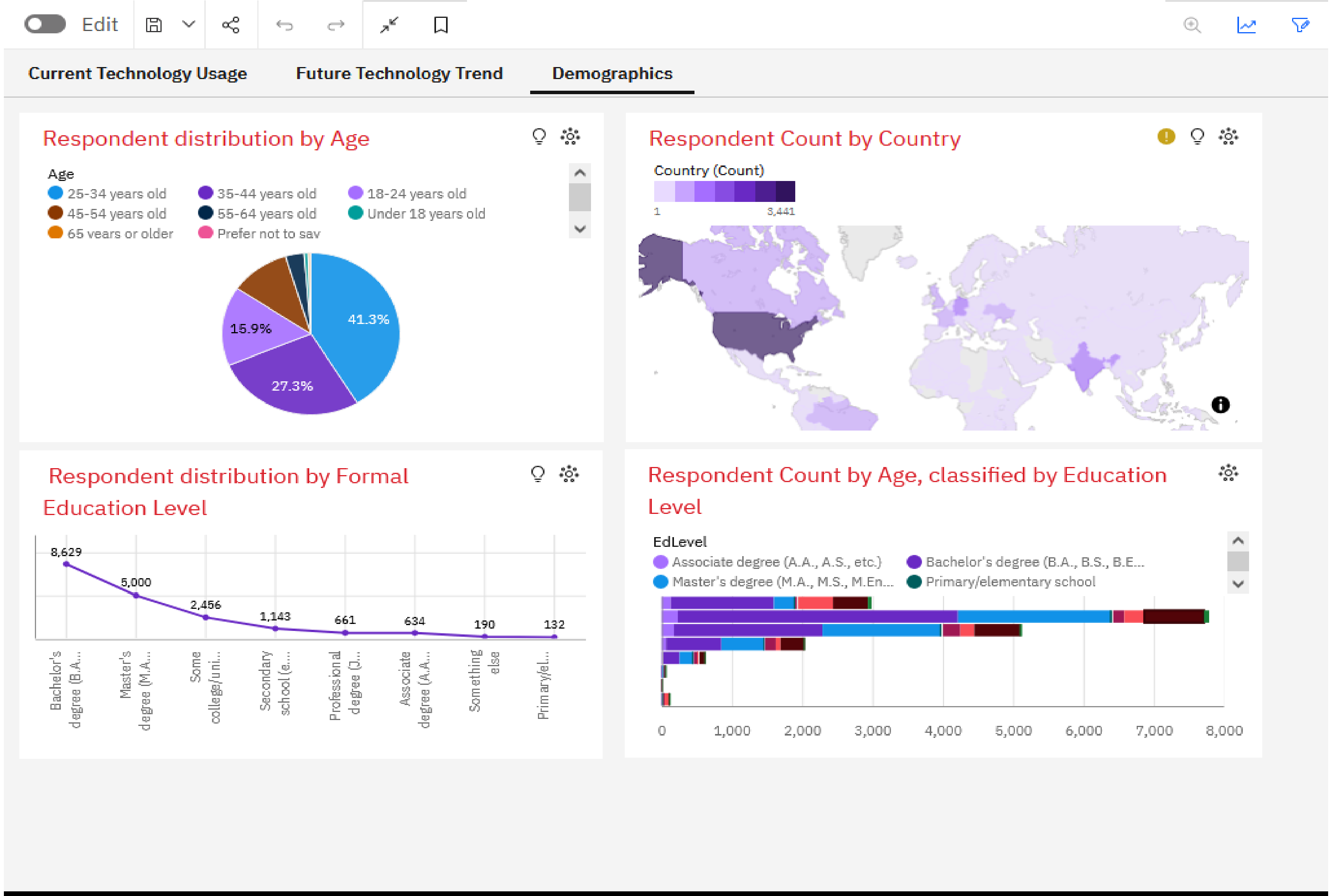
Future Technology Trend-Top 10 platforms most wanted to work with



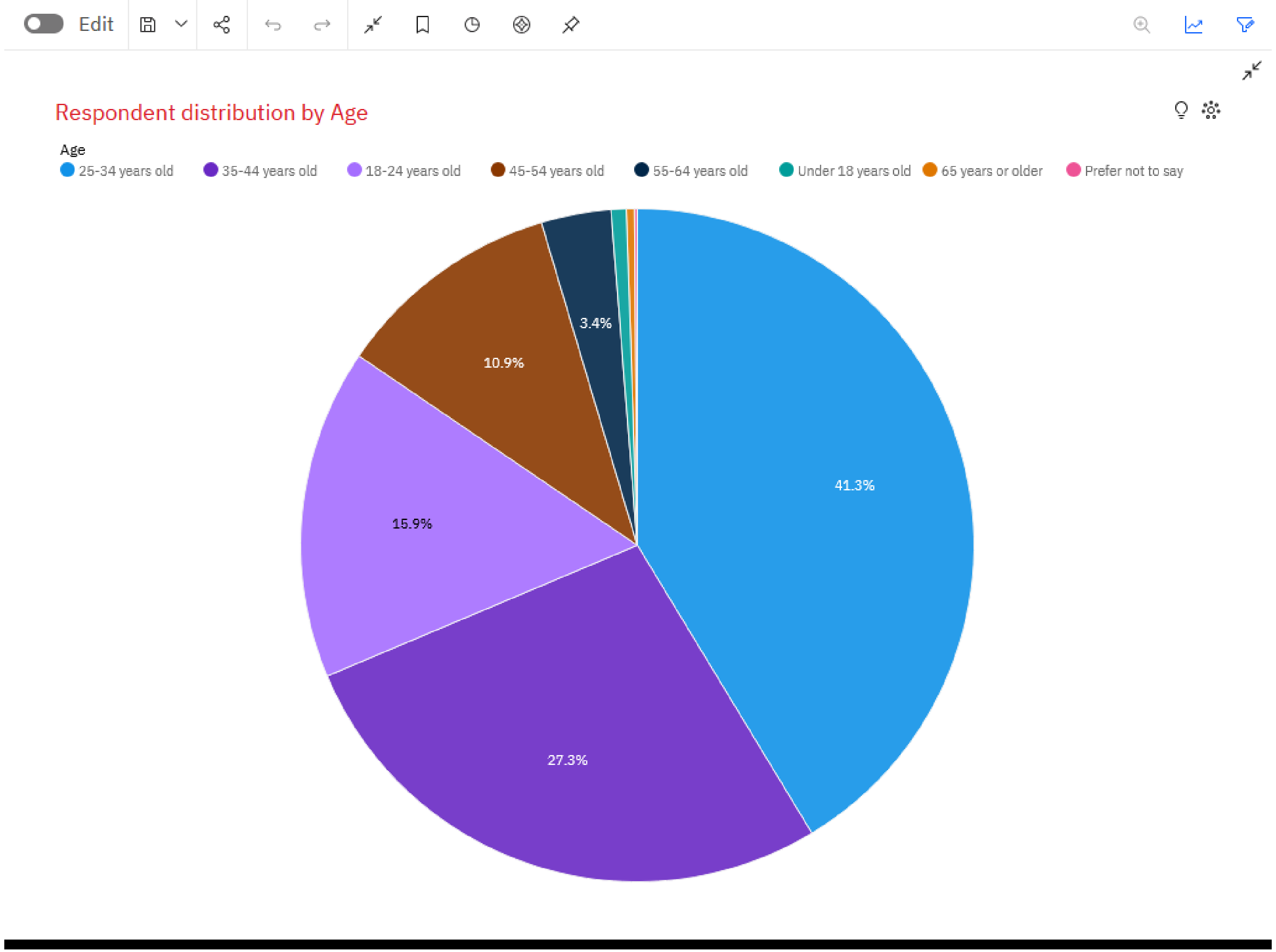
Future Technology Trend-Top 10 webframes most wanted to work with



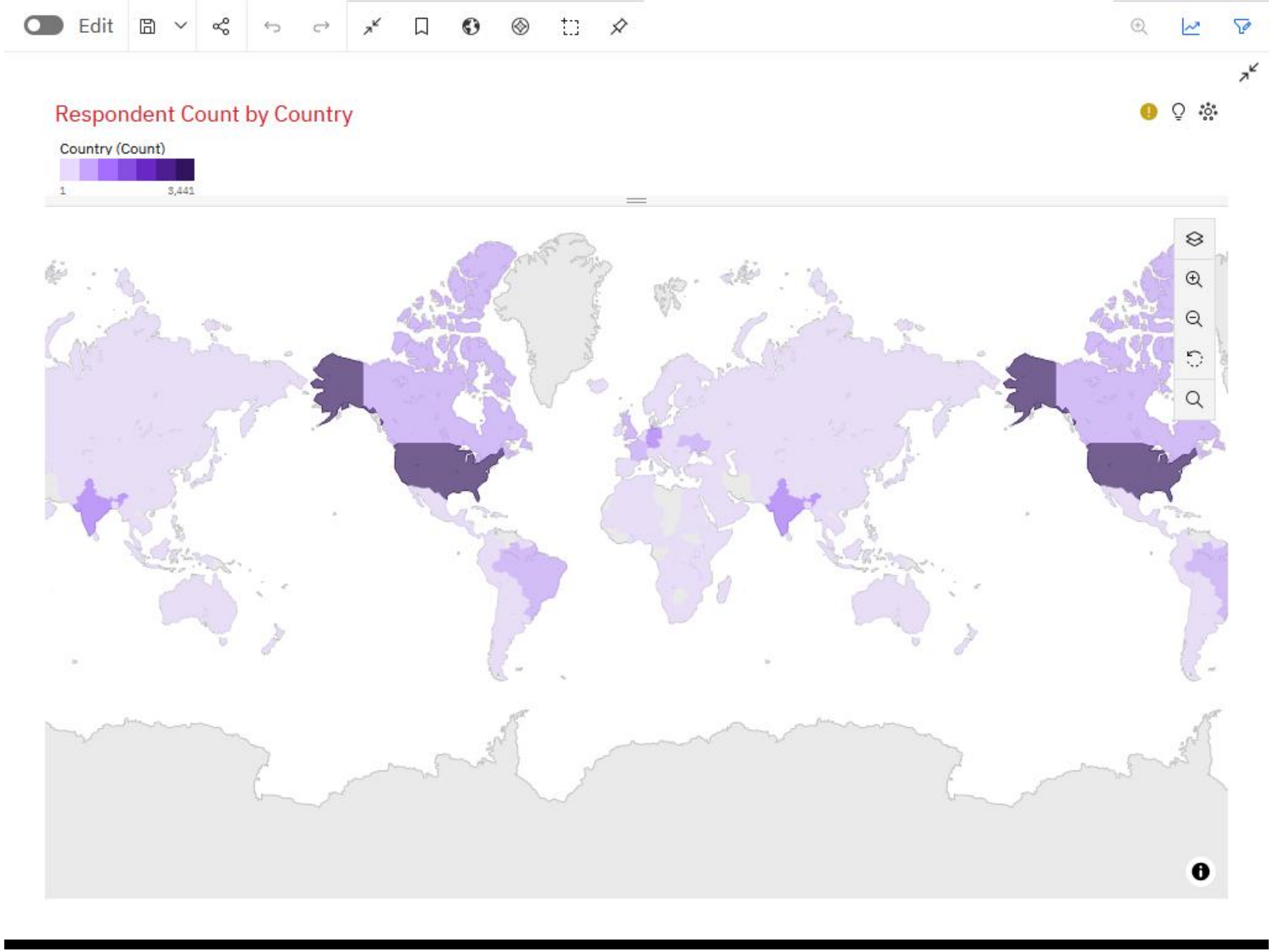
Demographics



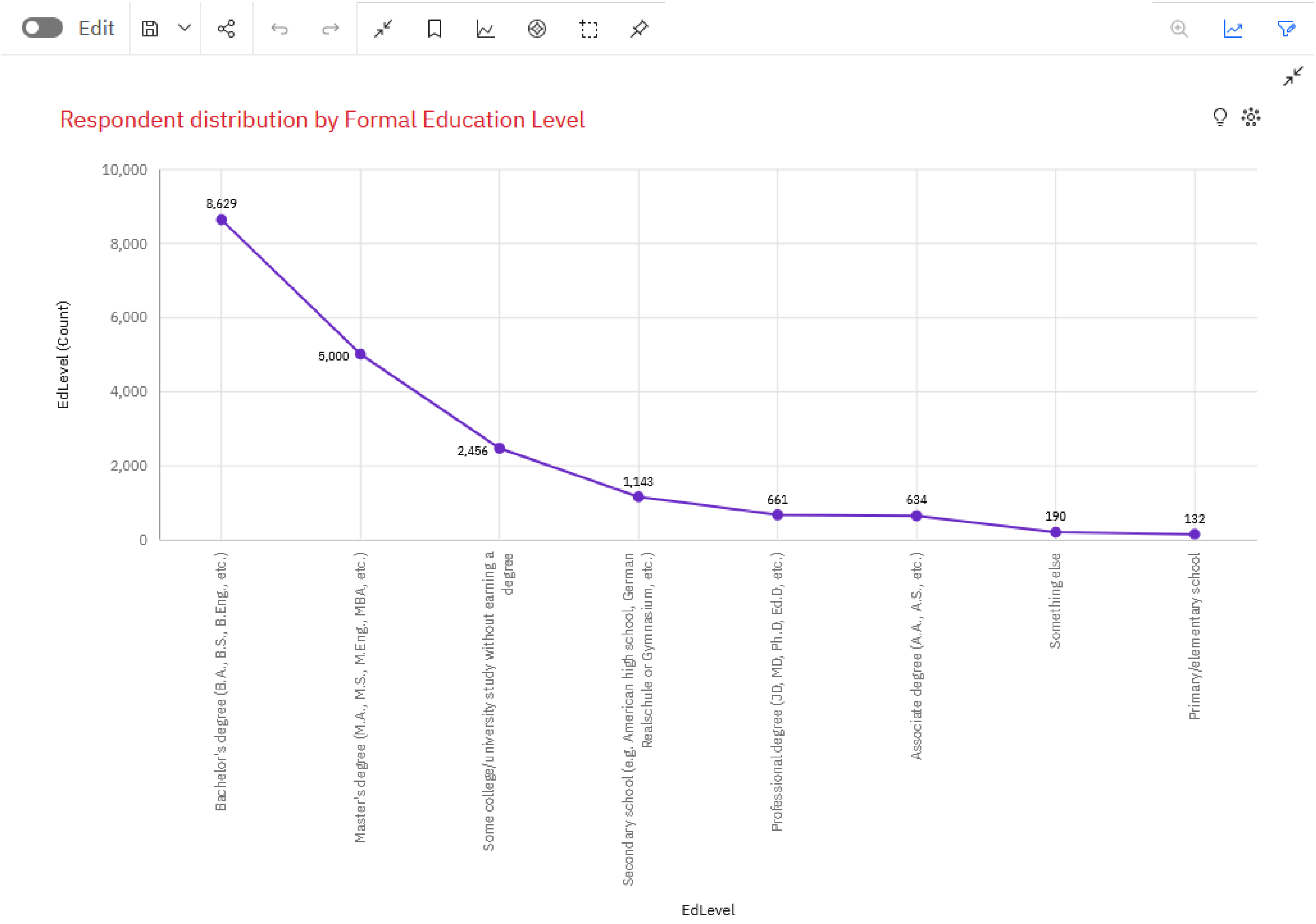
Demographics –Respondent distribution by Age



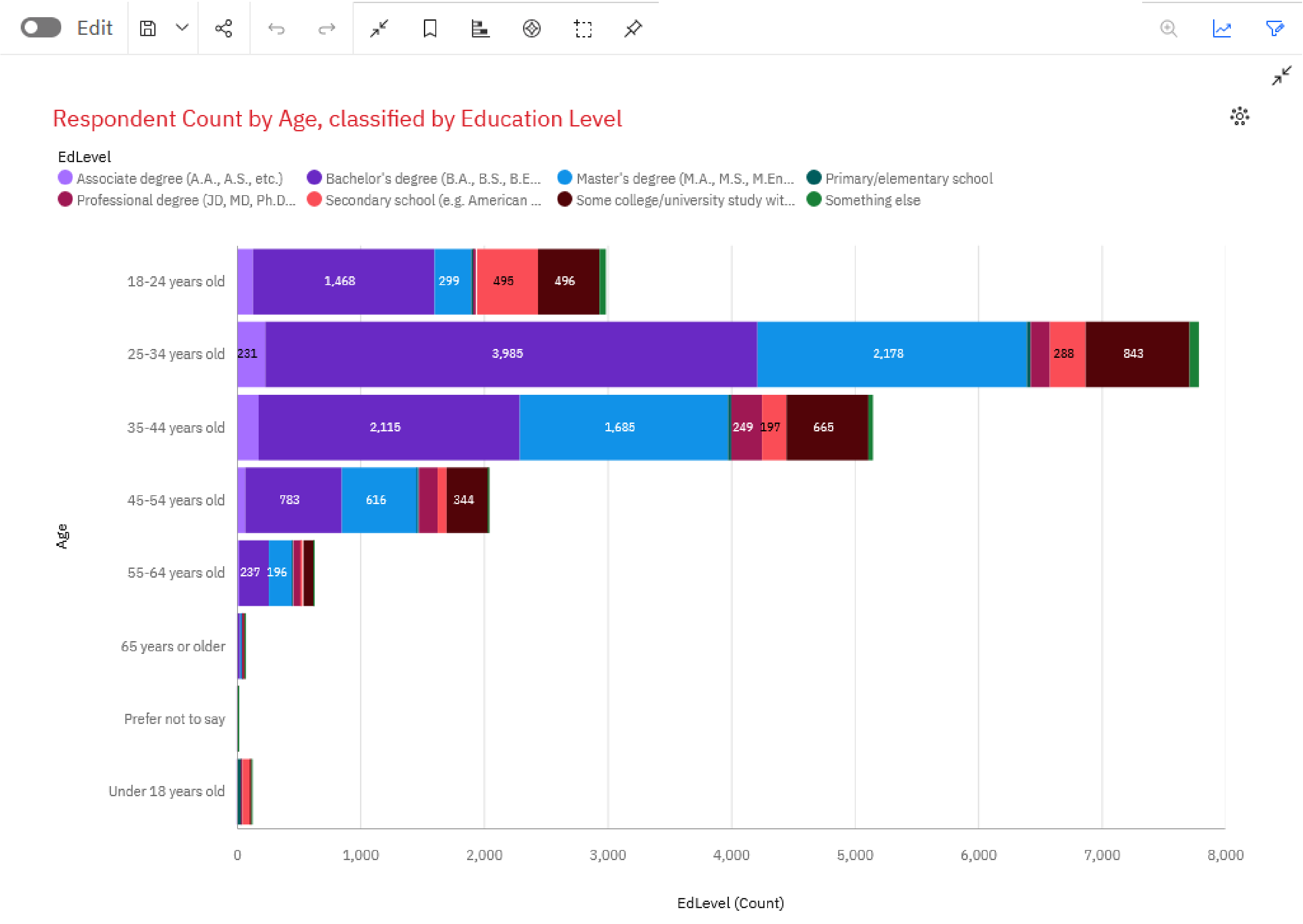
Demographics – Respondent count by Country



Demographics-Respondent distribution by Formal Education Level



Demographics-Respondent count by Age, classified by Education Level



DISCUSSION – DATA USED AND CHOICES MADE

- It is certain that if, instead of looking at a combination of languages, we had looked at each language separately, the result would have been quite different. In fact, it would have been interesting to ask respondents about their (single) most used language.
- Note: We could have extracted the languages individually and seen which one was the most popular. But this can be misleading and biased because a developer who says they use a combination of languages does not necessarily use a language from that combination more often; their most frequently used language on its own may be another language. For example, a full-stack developer may spend a lot of time on the front end (in which case they will use the HTML/CSS and JavaScript web stack) and a little less time on the back end with Java, for example. In this case, their most used language is actually the combination of the HTML/CSS and JavaScript web stack, whereas if they had to choose a single most used language, it would be Java.

DISCUSSION - RESULTS

- Current vs. Future: While JavaScript and Python lead today, the strong desire to work with AWS and Python points to a consolidation of cloud and data-centric skills.

- Demographic Insight: The primary workforce (25-34 years) is highly educated, suggesting a competitive job market where certifications and specialized skills (like cloud expertise) are key differentiators.

- High-Value Skills: Niche languages (Swift, Go) and cloud platforms (AWS) command premium salaries and high future demand.

OVERALL FINDINGS & IMPLICATIONS

Findings

- The tech landscape is bifurcating between widespread web technologies (JS, Python) and high-value specialized skills (Cloud, Swift, C++).
- Cloud proficiency is transitioning from a "nice-to-have" to a "must-have" skill.
- Real-world job postings sometimes contrast with developer surveys, highlighting the importance of analyzing multiple data sources.

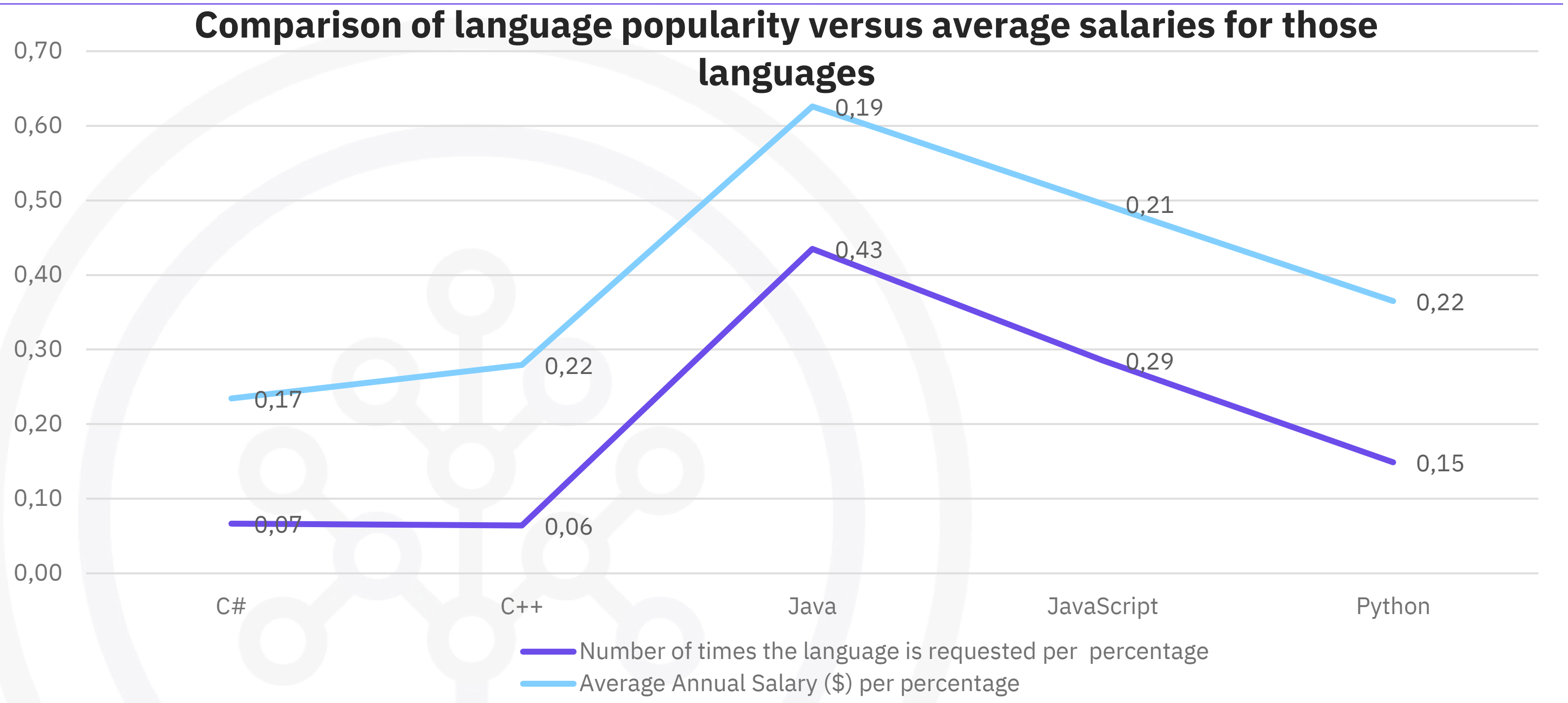
Implications

- For Job Seekers: Build a core foundation in Python and JavaScript, then specialize in a high-demand area like cloud computing or a niche language.
- For Educators: Curricula must integrate cloud technologies and data management principles.
- For Companies: To attract top talent, focus on projects involving trending technologies like AWS, Azure, Python, and Go.

CONCLUSION

- Python and JavaScript remain the cornerstone languages for the present and immediate future.
- Cloud Adoption is the single most significant trend, with AWS leading the pack.
- Data Skills—specifically SQL and cloud databases—are critically important across roles.
- Strategic Specialization in high-salary, high-demand technologies (e.g., Swift, Go, AWS) offers a strong career advantage.

APPENDIX

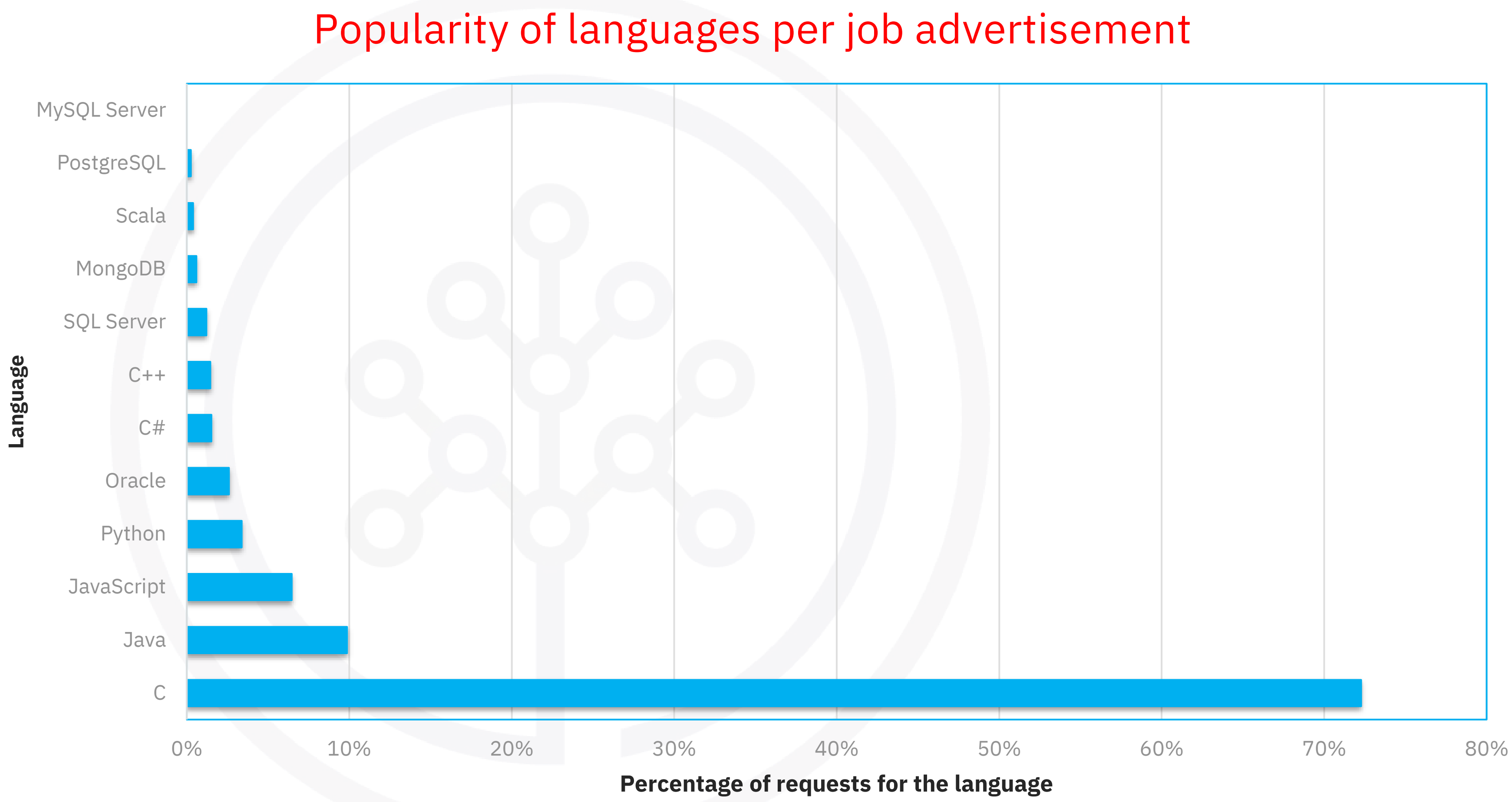


Language	Number of times the language is requested	Average Annual Salary ($)		Language	Number of times the language is requested per percentage	Average Annual Salary ($) per percentage
C#	526	$ 88 726		C#	0,07	0,17
C++	506	$ 113 865		C++	0,06	0,22
Java	3428	$ 101 013		Java	0,43	0,19
JavaScript	2248	$ 110 981		JavaScript	0,29	0,21
Python	1173	$ 114 383		Python	0,15	0,22
Total	7881	528968			1,00	1,00

We only considered languages that appeared in both job_postings.xlsx and popular-languages.xlsx. We used relative frequencies to enable comparison.

JOB POSTINGS

Language	Number of times the language is requested	Percentage of requests for the language in the sample
C	25114	72%
Java	3428	10%
JavaScript	2248	6%
Python	1173	3%
Oracle	899	3%
C#	526	2%
C++	506	1%
SQL Server	423	1%
MongoDB	208	1%
Scala	138	0%
PostgreSQL	86	0%
MySQL Server	0	0%
Total	34749	100%



POPULAR LANGUAGES

Language	Average Annual Salary
Swift	$ 130 801
Python	$ 114 383
C++	$ 113 865
Javascript	$ 110 981
Java	$ 101 013
Go	$ 94 082
R	$ 92 037
C#	$ 88 726
SQL	$ 84 793
PHP	$ 84 727